

UAV-Assisted Wireless Charging for Mobile Disaster IoT Networks: A Metaheuristic Optimization Approach

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Abstract—Maintaining network connectivity and situational awareness in disaster response scenarios requires robust Internet of Things (IoT) sensor networks capable of operating under dynamic, resource-constrained conditions. In this paper, we address the problem of scheduling a charging UAV in a Wireless Rechargeable Mobile Network (WRMN) deployed for disaster response applications, where Internet of Things devices (IoTDs) move continuously under the Gauss-Markov mobility model and must be kept operational to relay critical sensing data. We formulate UAV charging scheduling as an offline trajectory optimization problem and propose a mobility-aware composite fitness function based on mean normalized residual energy. We conduct a comprehensive comparative evaluation of four metaheuristic algorithms, Particle Swarm Optimization (PSO), Genetic Algorithm (GA), Grey Wolf Optimizer (GWO), and Whale Optimization Algorithm (WOA), under identical network conditions. Experimental results demonstrate that GA achieves the longest network lifetime (10.67 timeslots), highest mean residual energy (37.22 J), and best energy fairness index (0.936) among all evaluated algorithms, outperforming PSO by 14.2% in network lifetime. GWO achieves the lowest UAV energy consumption (583,940 J), offering a compelling efficiency trade-off for energy-constrained UAV platforms. These findings provide evidence-based algorithm selection guidance for UAV-assisted mobile IoT networks in disaster response deployments. Future work will explore multi-UAV extensions and online trajectory replanning to enhance the practical applicability of the proposed framework.

Index Terms—Wireless rechargeable sensor networks, UAV-assisted charging, trajectory optimisation, metaheuristic algorithms, network lifetime, Gauss-Markov mobility, disaster response, mobile IoT networks

I. INTRODUCTION

The Internet of Things (IoT) has become increasingly important in managing disasters and supporting critical tasks such as search and rescue, emergency tracking, and environmental monitoring [1]–[3]. However, during such events, ground-based power and communication infrastructure is often severely damaged or destroyed, leaving low-power sensing devices facing strict energy constraints that threaten continuous data collection. To address this, Unmanned Aerial Vehicles (UAVs) have emerged as flexible, cost-effective mobile energy transmitters for IoT devices forming Wireless Rechargeable

Mobile Networks (WRMNs) in hard-to-reach disaster environments [4], leveraging recent advances in wireless power transfer to recharge energy-depleted nodes where conventional infrastructure is unavailable [3], [5].

Disaster zones are constantly changing and filled with unknown barriers [5], [6], while devices are often attached to moving rescue workers or shifting assets, creating dynamic network topologies with continuous node mobility [4]. This requires the joint optimization of UAV flight trajectories, charging scheduling, and resource allocation under unpredictable node movement, a highly complex, multi-objective problem proven to be NP-hard [7], [8]. Yet most existing approaches assume static sensor networks, and metaheuristic algorithms [8], [9], uniquely capable of exploring massive solution spaces for such non-linear problems, remain underexplored for fully mobile WRSNs that must rapidly adapt to continuous IoT spatial variation.

In this work, we propose a mobility-aware UAV charging scheduling framework for mobile WRSNs, using a composite fitness function based on mean residual energy that implicitly maximizes network lifetime under node mobility. We evaluate four metaheuristics, PSO, GA, GWO, and WOA, demonstrating that GA achieves superior network-health metrics while GWO offers the most energy-efficient UAV operation, providing evidence-based algorithm selection guidance for UAV-assisted mobile IoT deployments. The remainder of this paper is organized as follows: Section II reviews related work; Sections III and IV present the system and energy models; Section V defines the problem formulation; Section VI discusses simulation results; and Section VII concludes the paper.

II. RELATED WORK

Recent literature has explored diverse UAV-assisted Wireless Power Transfer (WPT) scheduling algorithms for disaster environments. To address infrastructure loss, [6] developed a two-stage strategy combining Artificial Potential Fields (APF) with dynamic genetic clustering for trajectory optimization, while [4] introduced the "Age of Charging" (AoC) metric via

DDPG to recharge mobile IoT devices before energy depletion. For mission-critical reliability, [5] integrated multi-UAV swarms with Reconfigurable Intelligent Surfaces (RIS) through a hybrid DQN-PSO framework, and [2] proposed the AEADC scheme using air-to-air relay links to minimize data staleness from deep disaster zones. To combat environmental uncertainty, [10] applied distributionally robust optimization for weather-induced energy risks, [11] employed hybrid NSGA-II with multi-agent reinforcement learning to minimize deployed UAVs, and [3] used meta-reinforcement learning to adapt UAV swarm 3D trajectories in real-time as nodes frequently join or leave the network.

In UAV path planning and energy replenishment, population-based metaheuristics have been extensively applied to navigate complex, non-linear solution spaces. Genetic Algorithms (GA) have proven effective for spatial optimization, from improved GA for 3D hover location planning [12] to hybridization with reinforcement learning [13], fuzzy-logic path evolution [14], and NSGA-II for initializing reinforcement learning agents [11]. Particle Swarm Optimization (PSO) is favored for convergence speed, with multi-dimensional variants enabling joint scheduling and discrete trajectory optimization [15]. Bio-inspired alternatives further enhance search diversity: ACO with dynamic pheromone intensity for structural monitoring [16], Simulated Annealing (SA) to escape local minima during data collection [2], and the Black Hole (BH) algorithm demonstrating significantly faster execution than standard approaches for recharge scheduling [8].

III. SYSTEM MODEL

A. Network Model

We consider a Wireless Rechargeable Mobile Network (WRMN) comprising $N = 30$ IoT devices (IoTDs) deployed uniformly at random across a two-dimensional geographical area $\mathcal{R} = [0, L]^2$ of $100 \times 100 \text{ m}^2$, where $L = 100 \text{ m}$. Each IoTD is equipped with a rechargeable battery of maximum capacity $E_{\max} = 50 \text{ J}$. A single UAV is deployed as a mobile charging station to wirelessly replenish the energy of IoTDs throughout the operational horizon.

The UAV flies at a fixed altitude $h = 10 \text{ m}$ and its position at timeslot t is denoted $\Psi^t = (X^t, Y^t, h)$. The operational horizon spans $T = 20$ timeslots of equal duration $T_{\max} = 100 \text{ s}$. Within each timeslot t , the UAV travels from its current position to a designated charging location j , hovers to deliver wireless energy to IoTDs within charging range, and then transitions to the next timeslot. The timeslot budget imposes a hard constraint between flight time τ_m^t and hover duration τ_c^t :

$$T_{\max} = \tau_m^t + \tau_c^t, \quad \forall t = 1, 2, \dots, T. \quad (1)$$

The 3D Euclidean separation between the UAV and IoTD i at timeslot t is:

$$D_i^t = \sqrt{(X^t - x_i^t)^2 + (Y^t - y_i^t)^2 + h^2}, \quad (2)$$

where (x_i^t, y_i^t) are the 2D coordinates of IoTD i at timeslot t .

B. Node Mobility Model

IoTDs move continuously according to the Gauss-Markov mobility model [17], which captures temporal correlation in node velocity through a memory factor $\alpha \in [0, 1]$. At each timeslot t , the speed v_i^t and direction θ_i^t of IoTD i are updated as:

$$v_i^t = \alpha v_i^{t-1} + (1 - \alpha) \bar{v} + \sqrt{1 - \alpha^2} \sigma_v \eta_v^t, \quad (3)$$

$$\theta_i^t = \alpha \theta_i^{t-1} + (1 - \alpha) \bar{\theta} + \sqrt{1 - \alpha^2} \sigma_\theta \eta_\theta^t, \quad (4)$$

where $\bar{v}$ and $\bar{\theta}$ are the mean speed and mean direction, σ_v and σ_θ are the corresponding standard deviations, and $\eta_v^t, \eta_\theta^t \sim \mathcal{N}(0, 1)$ are independent Gaussian noise terms. The position of IoTD i is updated as:

$$x_i^t = x_i^{t-1} + v_i^t \cos(\theta_i^t) \cdot T_{\max}, \quad (5)$$

$$y_i^t = y_i^{t-1} + v_i^t \sin(\theta_i^t) \cdot T_{\max}. \quad (6)$$

Elastic boundary reflection is applied when a node reaches the boundary of $\mathcal{R}$. We adopt the standard parameter values $\alpha = 0.75$, $\bar{v} = 1 \text{ m/s}$, $v_{\max} = 2 \text{ m/s}$, and $\bar{\theta}$ uniformly initialised in $[0, 2\pi)$. Figure 1 shows an example of the considered network model.

IV. ENERGY MODELS

A. Wireless Power Transfer Model

The UAV delivers energy to IoTDs via electromagnetic (EM) resonance induction, a widely adopted WPT technology whose efficiency is insensitive to antenna misalignment and can exceed 90% at close range [18]. The charging efficiency η_c as a function of UAV-IoTD separation D_i^t is:

$$\eta_c = \frac{E_{\text{rec}}^{ij}}{P_{\text{Tx}}} = \frac{\vartheta}{\varsigma (D_i^t)^6 + 1}, \quad (7)$$

where P_{Tx} is the UAV transmit power, ϑ and ς are constant parameters determined by the hardware configuration, and D_i^t is the dominant governing factor. The energy received per second by IoTD i is thus $E_{\text{rec}}^{ij} = \eta_c P_{\text{Tx}}$.

The battery state of IoTD i at the end of timeslot t evolves as:

$$\tilde{E}_i^t = \min(E_i^t + (E_{\text{rec}}^{ij} - \xi_i^t) \tau_c^t, E_{\max}), \quad (8)$$

where ξ_i^t is the per-second energy consumption of IoTD i and τ_c^t is the hover duration. The $\min(\cdot, E_{\max})$ clamp prevents battery overflow. An IoTD is charged only if it lies within the effective charging range $D_{\max}$, defined as the maximum separation at which $E_{\text{rec}}^{ij} > \xi_i^t$, ensuring a positive net energy gain:

$$D_{\max} = \left(\frac{\vartheta P_{\text{Tx}}}{\varsigma \xi_i^t} - \frac{1}{\varsigma} \right)^{1/6}. \quad (9)$$

B. IoTD Energy Consumption Model

The per-second energy consumption rate ξ_i^t of IoTD i is modeled using the Heinzelman *et al.* first-order radio energy model [19]:

$$\xi_i^t = (1 + n_{\text{rx},i}^t) E_{\text{elec}} L + E_{\text{amp}} L (d_{\min,i}^t)^2, \quad (10)$$

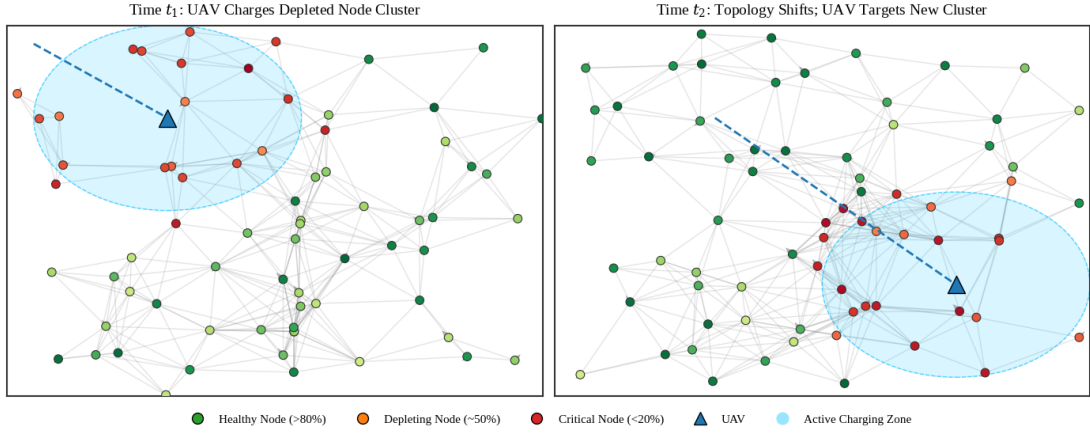


Fig. 1. Temporal evolution of a mobile wireless recharging IoT network, depicting a UAV performing dynamic energy harvesting on critical nodes before and after a network topology shift.

where E_{elec} is the electronics energy coefficient (transmit and receive circuitry), E_{amp} is the power amplifier coefficient, L is the packet size in bits, $d_{\min,i}^t$ is the distance from IoTD i to its nearest connected neighbour at timeslot t , and $n_{\text{rx},i}^t$ is the number of packets received per timeslot. We adopt the standard values $E_{\text{elec}} = 50 \text{ nJ/bit}$, $E_{\text{amp}} = 100 \text{ pJ/bit/m}^2$, and $L = 2000 \text{ bits}$ [19]. IoTDs that are isolated (no neighbour within the transmission range R_{Tx}) incur a worst-case transmission cost using $d_{\min,i}^t = L$.

C. UAV Energy Consumption Model

The UAV energy consumed during timeslot t is computed using the empirical quadrotor model of Abeywickrama *et al.* [20]. Since the UAV operates at a fixed altitude h throughout, vertical ascent and descent terms vanish, and the model reduces to:

$$E_{\text{uav}}^t = \underbrace{29.027 t_\alpha - 0.087}_{\text{arming}} + \underbrace{-0.432 V^2 + 3.786 V - 1.224}_{\text{take-off}} + \underbrace{308.709 \tau_m^t - 0.852}_{\text{horizontal flight}} + \underbrace{(4.917 h + 275.204) \tau_c^t}_{\text{hover}}, \quad (11)$$

where t_α is the arming stabilization duration, V is the take-off speed, and τ_m^t and τ_c^t are the horizontal flight and hover durations per timeslot, respectively. All empirical coefficients are taken directly from [20]. The UAV cruises at a constant speed $\nu \leq \nu_{\text{max}}$, and the flight time is therefore $\tau_m^t = \|\Psi^t - \Psi^{t-1}\|_2 / \nu$.

V. PROBLEM FORMULATION

A. Solution Encoding

With fixed UAV altitude, the decision variables per timeslot reduce to the 2D charging location and the hover duration. A candidate solution $\mathbf{x}$ encodes the full T -timeslot UAV dispatch policy as:

$$\mathbf{x} = \{(X_j^t, Y_j^t, \tau_c^t)\}_{t=1}^T, \quad (12)$$

yielding a real-valued vector of dimension $3T = 60$. All remaining per-timeslot quantities, including UAV-IoTD separations D_i^t , received energies E_{rec}^{ij} , updated battery states $\tilde{E}_i^t$, and UAV energy expenditure E_{uav}^t , are derived deterministically through forward simulation of the models in Section III, given the IoTD trajectories generated by the Gauss-Markov model. The boundary constraints $0 \leq X_j^t \leq L$, $0 \leq Y_j^t \leq L$, and $\tau_c^t \leq T_{\text{max}} - \|\Psi^t - \Psi^{t-1}\|_2 / \nu_{\text{max}}$ are enforced by clamping during initialization and after each position update.

B. Optimization Objective

The network lifetime T_{life} is defined as the first timeslot at which any IoTD's residual energy falls below the minimum operational threshold Γ_E :

$$T_{\text{life}} = \min\{t \in \{1, \dots, T\} : \exists i \text{ s.t. } \tilde{E}_i^t < \Gamma_E\}, \quad (13)$$

with $T_{\text{life}} = T$ if all IoTDs remain operational throughout the horizon. Maximizing T_{life} is the primary goal. The scheduling optimization problem is stated as:

$$(P1) : \underset{\mathbf{x}}{\text{maximise}} \quad \mathcal{F}(\mathbf{x}) \quad (14a)$$

$$\text{s.t.} \quad 0 \leq X_j^t \leq L, \quad 0 \leq Y_j^t \leq L, \quad (14b)$$

$$\tau_c^t \leq T_{\text{max}} - \tau_m^t, \quad (14c)$$

$$D_i^t \leq D_{\text{max}}, \quad (14d)$$

$$\nu \leq \nu_{\text{max}}, \quad (14e)$$

$$\tilde{E}_i^t \geq \Gamma_E, \quad \forall i, t. \quad (14f)$$

C. Fitness Function

We define a composite fitness function that provides a smooth, continuous signal while implicitly maximizing network lifetime. The primary term is the *mean normalized residual energy*, averaged across all IoTDs and all timeslots:

$$\Phi(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^T \frac{1}{N} \sum_{i=1}^N \frac{\tilde{E}_i^t}{E_{\text{max}}}. \quad (15)$$

This metric is always in $[0, 1]$, is smooth with respect to the decision variables, and implicitly captures network lifetime: a

TABLE I
SIMULATION AND ALGORITHM PARAMETERS

Category	Parameter	Symbol	Value
Network	Area	L	$100 \times 100 \text{ m}^2$
	No. of IoTDs	N	30
	Max battery	$E_{\max}$	50 J
	Min. threshold	Γ_E	5 J
	Tx range	R_{Tx}	20 m
Mobility	Memory factor	α	0.75
	Mean speed	$\bar{v}$	1 m/s
	Max speed	$v_{\max}$	2 m/s
UAV	Altitude	h	10 m
	Max speed	$v_{\max}$	10 m/s
	Timeslot duration	$T_{\max}$	100 s
	Horizon	T	20
Algorithms	Population size	M	30
	Max iterations	$G_{\max}$	200
	Penalty coeff.	μ	1.0
	Independent runs	—	30
PSO	Inertia weight	w	$0.9 \rightarrow 0.4$
	Learning factors	c_1, c_2	2.0, 2.0
GWO/WOA	Decay parameter	a	$2 \rightarrow 0$
	Spiral constant	b	1.0

node that dies at timeslot t^* contributes $\tilde{E}_i^t = 0$ to the sum for all $t \geq t^*$, imposing a sustained penalty proportional to the premature death duration. This ensures that every node at every timeslot contributes equally regardless of spatial density, naturally directing the UAV toward isolated or depleted nodes that have drifted away from the network center. A normalized penalty term is added to enforce the energy floor constraint (14f):

$$\mathcal{P}(\mathbf{x}) = \frac{\mu}{NT} \sum_{t=1}^T \sum_{i=1}^N \max\left(0, \frac{\Gamma_E - \tilde{E}_i^t}{E_{\max}}\right), \quad (16)$$

where $\mu > 0$ is the penalty coefficient. Dividing by NT keeps the penalty on the same $[0, 1]$ scale as $\Phi(\mathbf{x})$, eliminating the need for sensitivity tuning as N or T vary. Setting $\mu = 1$ is sufficient as a default since both terms are normalized. We evaluate four population-based metaheuristic algorithms on problem (P1) under identical conditions. The algorithms are PSO [21], GA [22], GWO [23], and WOA [24]. Table I summarizes all simulation and algorithm parameters.

VI. RESULTS AND DISCUSSION

This section presents simulation results from 30 independent runs of each algorithm under identical mobile WRSN conditions, as described in Section III-A.

Fig. 2(a) shows best-fitness convergence curves averaged across all runs. GA achieves the highest final fitness (0.743 ± 0.013) and converges most rapidly, reaching $\sim 95\%$ of its terminal value within 150 iterations. PSO converges competitively (0.708 ± 0.007) but with a slower initial climb, reflecting premature convergence tendencies under mobility-induced fitness landscape shifts. GWO (0.698 ± 0.018) and WOA (0.691 ± 0.018) trail throughout, with all four curves remaining upward-trending at iteration 300, suggesting further gains are achievable with additional computational budget.

Fig. 2(b) shows network lifetime T_{life} . GA achieves the highest mean of 10.67 ± 1.25 timeslots, followed by WOA

(10.33 ± 1.89), GWO (10.00 ± 0.82), and PSO (9.33 ± 0.47), with GA extending network survival by 14.3% over PSO. The consistent gap between all algorithms and the $T = 30$ red dashed line reflects the inherent difficulty of protecting 30 mobile IoTDs with a single resource-constrained UAV, confirming that the proposed formulation is sufficiently challenging to meaningfully differentiate algorithm performance.

The dead-node timeline in Fig. 2(c) reveals temporal dynamics not captured by scalar lifetime metrics. PSO and GA show controlled accumulation, with fewer than three simultaneous deaths at any timeslot, whereas GWO and WOA exhibit sharp spikes peaking at approximately eight dead nodes around timeslot 10. This oscillatory pattern is consistent with their distributed-leadership search behavior, which periodically commits the UAV to distant hover locations, temporarily abandoning nearby depleted nodes. The total dead-node timeslots (TDNT) metric in Table II quantifies this effect: GA accumulates the fewest (17.67 ± 6.94), while GWO accumulates the most (40.67 ± 10.87), nearly $2.3\times$ worse.

In terms of mean residual energy (Fig. 2(d)), all algorithms maintain 31–35 J (62–70% of $E_{\max} = 50$ J), with GA achieving the highest MRE (37.22 ± 0.62 J). The small inter-algorithm differences suggest all approaches deliver energy effectively on average, but differ in distribution equity. As shown in Fig. 2(e), GA achieves the highest Jain fairness index (0.936 ± 0.031), approaching near-uniform energy distribution, while WOA scores lowest (0.841 ± 0.005), indicating a tendency to concentrate charging on a subset of nodes. Examining the per-timeslot energy profile in Fig. 2(f), PSO and GA sustain mean energies consistently above 15 J, whereas GWO and WOA exhibit pronounced oscillations during long UAV transit slots. Notably, all algorithms remain above $\Gamma_E = 5$ J throughout, confirming that the proposed fitness function successfully sustains network-wide energy levels.

Trajectory snapshots at $t \in \{1, 5, 10, 15, 20, 30\}$ are shown in Fig. 3 under identical network conditions. By $t = 5$, GA and PSO already demonstrate more spatially distributed charging compared to the localized GWO and WOA patterns. Dead nodes (x) begin appearing by $t = 10$ in the PSO, GWO, and WOA rows, concentrated at the deployment periphery where UAV charging efficiency is lowest. The darkening UAV trail clearly differentiates each algorithm's spatial strategy: GA develops a broad sweeping pattern covering more of the $100 \times 100 \text{ m}^2$ area, while WOA makes long diagonal transits that sacrifice hover duration for travel. By $t = 30$, GA maintains the fewest dead nodes across all runs. Table II summarizes all key performance metrics.

VII. CONCLUSION

This paper addressed UAV charging scheduling in Wireless Rechargeable Mobile Networks (WRMNs) for disaster response, where IoT devices move continuously under the Gauss-Markov mobility model. We formulated the problem as an offline trajectory optimization task and proposed a mobility-aware composite fitness function based on mean normalized residual energy that implicitly maximizes network

Performance Comparison of PSO, GA, GWO, and WOA
on Mobile WRSN UAV Charging Scheduling

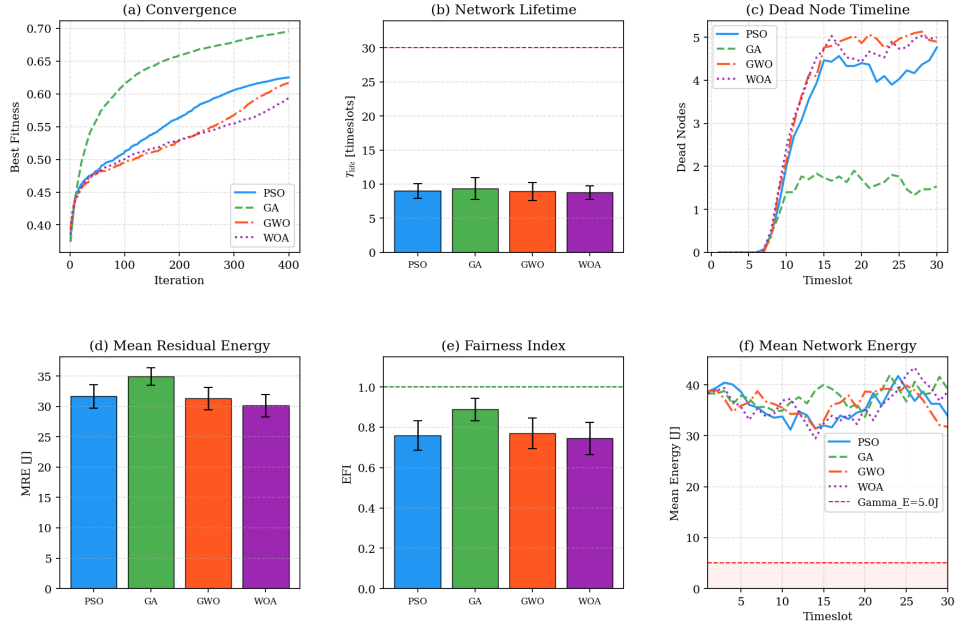


Fig. 2. Performance comparison of PSO, GA, GWO, and WOA across 30 independent runs: (a) convergence curves, (b) network lifetime, (c) dead-node timeline, (d) mean residual energy, (e) Jain's fairness index, and (f) mean network energy over the 30-timeslot horizon. Red dashed line in (b) denotes the full horizon; shaded region in (f) marks the energy death zone below $\Gamma_E = 5$ J.

TABLE II
PERFORMANCE COMPARISON OF METAHEURISTIC ALGORITHMS ACROSS 30 INDEPENDENT RUNS (MEAN $\pm$ STD). BEST VALUE PER METRIC SHOWN IN BOLD.

Algorithm	T_{life} [slots]	MRE [J]	EFI	TDNT	Fitness	E_{UAV} [kJ]	CCR	Conv. Iter
PSO	8.00 ± 0.00	32.79 ± 1.50	0.780 ± 0.094	57.0 ± 5.4	0.650 ± 0.030	766.0 ± 34.6	1.000 ± 0.000	392.0 ± 5.0
GA	8.00 ± 0.00	33.25 ± 1.05	0.827 ± 0.031	75.0 ± 5.7	0.658 ± 0.021	694.8 ± 42.8	1.000 ± 0.000	382.7 ± 9.5
GWO	9.00 ± 0.82	30.84 ± 1.26	0.815 ± 0.080	110.3 ± 27.3	0.606 ± 0.028	563.5 ± 19.7	1.000 ± 0.000	395.0 ± 0.8
WOA	8.00 ± 0.00	29.24 ± 1.99	0.686 ± 0.026	113.0 ± 42.4	0.574 ± 0.044	667.3 ± 78.8	0.989 ± 0.016	397.3 ± 0.5

T_{life} : network lifetime. MRE: mean residual energy. EFI: Jain's fairness index ($\uparrow$ better). TDNT: total dead-node timeslots ($\downarrow$ better).

E_{UAV} : UAV energy consumption ($\downarrow$ better). CCR: charging coverage rate ($\uparrow$ better). Conv. Iter: convergence iteration ($\downarrow$ better).

lifetime. Four metaheuristic algorithms, PSO, GA, GWO, and WOA, were evaluated over 30 independent runs on a 30-node, $100 \times 100 \text{ m}^2$ mobile WRSN. GA achieved the highest fitness (0.658 ± 0.021), mean residual energy (33.25 ± 1.05 J), and fairness index (0.827 ± 0.031), while GWO attained the longest network lifetime (9.00 ± 0.82 timeslots) and lowest UAV energy consumption (563.5 ± 19.7 kJ). Trajectory analysis confirmed that GA develops broader spatial coverage patterns servicing peripheral nodes more consistently, whereas GWO and WOA exhibit oscillatory behavior with periodic energy drops during long transit flights. Per-node energy analysis further revealed that node drift to the network periphery remains the principal scheduling challenge across all algorithms. Future work will extend this framework to online replanning with per-timeslot policy updates using true observed node positions, and to multi-UAV coordination for larger-scale disaster response deployments.

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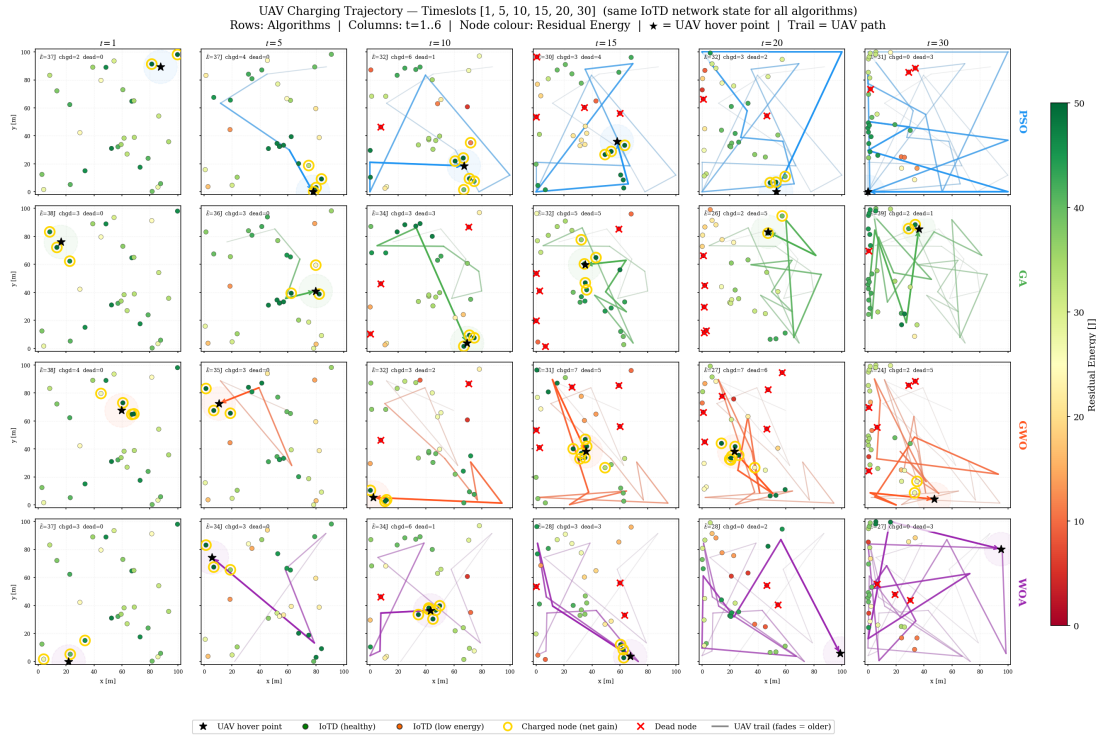


Fig. 3. UAV charging trajectory snapshots at timeslots $t \in \{1, 5, 10, 15, 20, 30\}$ for all four algorithms (rows) evaluated on an identical IoT network assumptions. Node color indicates residual energy (green = full, red = critical); gold rings denote nodes receiving a net energy gain at that timeslot; red crosses mark dead nodes. The UAV trail darkens from grey to the algorithm's color as timeslots progress, and the star marks the current hover location.

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